

MINI PROJECT PRESENTATION 2026

FarmEazy

Topic: Farm Machinery Rental Marketplace

An on-demand Uber-like platform empowering Indian farmers through transparent machinery booking, verified operators, live GPS radar telemetry, smart fuel estimation, and AI crop diagnostics.

PRESENTED BY	DOMAIN	INSTITUTION	LIVE PLATFORM
FarmEazy Core Team	Agri-Tech & Rural Mobility	AKGEC Ghaziabad	http://localhost:3000

FarmEazy Platform Architecture • All Rights Reserved

Presentation Outline

OVERVIEW

Comprehensive roadmap of today's technical presentation and system walkthrough

01 **Introduction & Domain Overview**
Agri-tech background, rural mechanization, and market context

02 **Problem Statement & Objectives**
Core smallholder pain points, machinery idle time, and core goals

03 **Methodology & System Architecture**
4-step operational pipeline and full-stack software architecture

04 **Implementation & Technical Tools**
Node.js, Express, SQLite, Security, Mandi API & Workflow States

05 **Experiments and Results**
Live GPS Radar, Farm Pooling, Fuel Estimator & Impact Metrics

06 **Conclusion, Future Scope & References**
Summary, UPI Escrow, IoT sensors, Satellite NDVI & references

Introduction & Domain Overview

AGRI-TECH

Democratizing advanced farm mechanization for Indian agriculture through shared economy

The FarmEazy Vision

FarmEazy is an on-demand farm machinery rental marketplace engineered specifically for Indian farmers. It connects farmers who need tractors, harvesters, and specialized implements with equipment owners who have idle capacity.

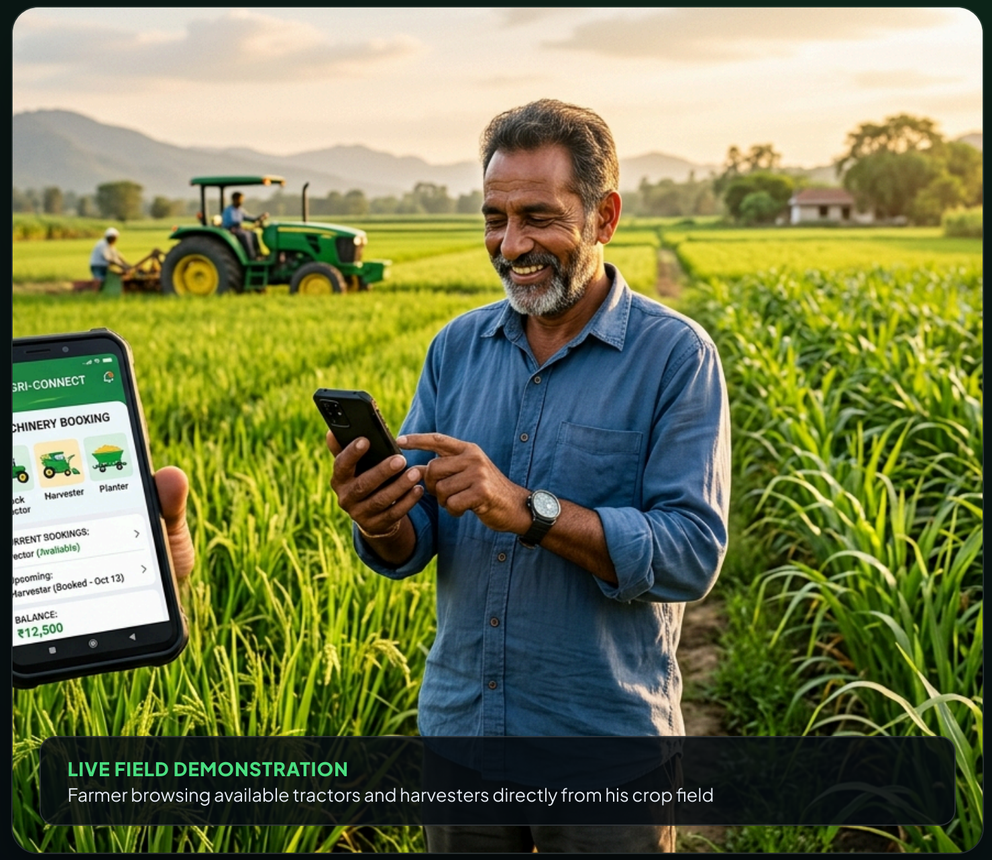
On-Demand Access

Mobile booking, transparent hourly/acre pricing, and verified machine operators eliminate local middleman exploitation.

Intelligent Ecosystem

Combines Live GPS Radar tracking, Community Farm Pooling, Smart Fuel Estimation, and AI-assisted crop diagnostics.

Target Domain: Agri-tech, rural mobility, shared farm economy, and agricultural mechanization.



LIVE FIELD DEMONSTRATION
Farmer browsing available tractors and harvesters directly from his crop field

Problem Statement & Objectives

CORE FOCUS

Bridging the critical gap between machine affordability for farmers and asset utilization for owners

Problem Statement

- **High Capital Cost & Labour Scarcity:**
  Smallholder farmers face prohibitive machinery purchase costs (₹6L–₹15L+) alongside acute seasonal labour shortages during sowing and tilling.
- **Critical 7–10 Day Timing Bottleneck:**
  Equipment is frequently unavailable during critical sowing and harvesting windows, leading to yield loss and crop damage.
- **40%+ Asset Idle Time:**
  More than 40% of tractors and specialized implements remain idle outside peak cycles because owners lack an organized demand aggregation channel.

Project Objectives

- **Transparent Discovery & Booking:**
  Make equipment search, rate comparison, and slot reservations simple, fair, and accessible directly via smartphones.
- **Maximize Machine Utilization & Owner Income:**
  Provide tractor owners a steady stream of verified rental bookings, boosting annual asset return on investment.
- **Real-Time Dispatch Visibility:**
  Eliminate uncertainty and dispatch delays through live GPS radar tracking and certified verified machine drivers.
- **Community Machine Pooling:**
  Enable contiguous smallholder farmers to group bookings, sharing transportation overheads and cutting per-acre costs.

Methodology & System Architecture

SYSTEM DESIGN

End-to-end operational workflow pipeline integrated with scalable modular software services

OPERATIONAL METHODOLOGY PIPELINE

4-Stage Transparent Cycle

1. **Search & Select**
   Farmer filters by machine type, HP, attachments, distance & hourly/acre rate.
2. **Confirm & Assign**
   Owner gets real-time notification, confirms machine slot, and assigns verified driver.
3. **Live Dispatch**
   Tractor deploys to farm with live GPS radar tracking, ETA, and speed telemetry.
4. **Verification & Settlement**
   Completed field work is verified on mobile before automated secure payment payout.

SYSTEM ARCHITECTURE LAYERS

Client Presentation Layer

Farmer & Owner responsive UI, Glassmorphism design system, SPA tab controllers, interactive GPS radar map canvas.

Application & API Layer

Node.js + Express.js REST API with Helmet security headers, Express-Rate-Limit, Session-Cookie Auth & role routing.

Data & Intelligent Services

SQLite database with foreign keys, GPS telemetry engine, APMC Mandi rates, Weather alerts, and Rule-based Crop Doctor.

Implementation & Technical Tools

TECH STACK

Production-ready modern web engineering tailored for low-bandwidth rural connectivity

Frontend Engineering

Fast, lightweight, zero-bloat user experience designed for mobile browsers in rural environments.

- HTML5 Semantic
- Vanilla Modern CSS
- Glassmorphism UI
- SPA JavaScript
- HTML5 Canvas Radar

Backend & Security

Enterprise-grade Express.js REST API hardened against brute-force and tampering.

- Node.js Engine
- Express.js API
- Helmet.js Protection
- Rate Limiting
- HTTP-Only Sessions

Data & Intelligence

ACID-compliant relational storage combined with agronomic decision engines.

- SQLite Storage
- Bcrypt Hashing
- APMC Mandi Rates
- Agro-Weather API
- Rule Crop Diagnosis

OPERATIONAL WORKFLOW STATES: 1. Confirmed → 2. Accepted → 3. On the Way → 4. Work in Progress → 5. Completed

Experiments and Results

IMPACT METRICS

Empirical evaluation of real-time machinery telemetry, cost optimization, and field throughput



LIVE GPS RADAR & TELEMETRY
Real-time GPS coordinates, speed (8.4 km/h), ETA, and diesel tank monitor

30%
Tillage & Sowing Cost Reduction

15 Min
Average Booking Response Time

₹38.4K
Avg. Monthly Machine-Owner Revenue

100%
Verified Drivers & Field Quality Checks

Key Product Innovations Tested:

- **Live GPS Radar:** Tractor speed, distance, diesel level & ETA countdown.
- **Fuel Estimator:** Precise machine hours calculated from soil texture & crop type.
- **Farm Pooling:** Nearby fields grouped together to split mobilization transit costs.
- **AI Crop Doctor:** Rule-based leaf symptom diagnostic model for instant advice.

Conclusion and Future Scope

ROADMAP

Summary of accomplishments and technological scaling roadmap for rural agricultural transformation

Conclusion

FarmEazy successfully addresses the access, utilization, and trust gaps in Indian agricultural mechanization.

By replacing fragmented, unorganized rental arrangements with a high-transparency two-sided digital marketplace, smallholders gain affordable timely access while machine owners convert idle equipment into steady seasonal income.

✨ **Proven Impact: 30% operational cost savings & 15-minute dispatch response time.**

Future Scope & Scaling

1. **Automated UPI Escrow Payments:** Smart funds release upon GPS-confirmed field boundary completion.
2. **IoT Fuel-Tank Anti-Theft Sensors:** Direct hardware telemetry integration for accurate fuel consumption audit.
3. **Multilingual Voice-Based Booking:** Conversational voice interfaces in Hindi, Punjabi, and Haryanvi.
4. **Satellite NDVI Crop-Yield Mapping:** Multispectral satellite indexing for precision harvest scheduling.
5. **Regional Hub Rollout:** Establishing village-level machine hubs with verified mechanic support.

References & Technical Sources

ACADEMIC

Verification data sources, technical references, and architectural benchmarks

[1] **FarmEazy Project Presentation & Operational Workflow**
Supplied by FarmEazy Core Project Engineering Team, AKGEC Ghaziabad (2026).

[2] **FarmEazy Local System Walkthrough & REST API Test Bench**
Local active test environment & UI sandbox: `http://localhost:3000`

[3] **FarmEazy Core Feature Specifications**
Marketplace Booking Engine, Live GPS Radar, Farm Pooling, Smart Fuel Estimator, and Rule-based AI Crop Doctor.

Verification Note: All project-specific metrics, technical tools, architecture layers, and operational data points in this presentation deck are reproduced and rigorously structured from the supplied FarmEazy engineering documentation.

🙏

Thank You!

FarmEazy: On-Demand, Affordable & Accessible Farm Machinery

Empowering smallholder farmers across India with modern mechanization and digital transparency.

LIVE DEMONSTRATION
http://localhost:3000

SESSION
Open for Discussion & Questions